

9.1-9.4

9.1

three dimensional operators are just the products of the one dimensional case, since each direction is orthogonal. Instead of single integrals we have triple, and bras/kets are now expressed as

$$\langle x, y, z | \psi \rangle$$

We can express things radially, and with gradients now.

9.2

We have

$$\hat{\mathbf{p}}_1^2 = \hat{p}_{1x}^2 + \hat{p}_{1y}^2 + \hat{p}_{1z}^2$$

Inversion is a discrete symmetry (parity operator)

9.4

smaller atoms have higher kinetic energy, so more natural resistance to compressing the atom in a small area?